

Yang-Mills Mass Gap $\Delta_{YM}(T)$ via $S_{26}^{\{(k)\}}$

Daniel T. Murphy

PAPER_971: Yang-Mills Mass Gap $\Delta_{YM}(T)$ via $S_{26}^{\{(k)\}}$

Author: Daniel T. Murphy — Star Magic / UQFF Framework

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Source: qgp_ramanujan_application.py (YangMillsMassGapCalculator)

Calculator: YangMillsMassGapCalc (CP4 #555)

CVW: v2.0.0 compliant

Abstract

We express the Yang-Mills mass gap $\Delta_{YM}(T)$ in terms of the QCD scale Λ_{QCD} and the expanded Ramanujan factor $S_{26}^{\{(k)\}}$. The gap closes continuously at the deconfinement temperature $T_c = 1.5 \times 10^{12}$ K, providing a UQFF-native derivation of confinement/deconfinement.

1. Yang-Mills Mass Gap

$$\Delta_{YM}(T) = \Lambda_{QCD} \cdot \left(1 - \frac{T}{T_c}\right) \cdot S_{26}^{\{(k)\}} / ([SSq])$$

For $T < T_c$: gap is open (confinement). At $T = T_c$: gap closes (deconfinement).

2. Connection to Millennium Prize

The Clay Millennium Prize Yang-Mills problem asks for proof that gauge theory has a mass gap $\Delta > 0$. The UQFF framework provides:

- A temperature-dependent gap function
- Closure at T_c via $S_{26}^{\{(k)\}}$ modulation
- Consistency with lattice QCD measurements of $\Lambda_{QCD} \approx 217$ MeV

3. Gap vs Temperature

T (K)	Δ_{YM} (eV)	State
10^{11}	$\approx \Lambda_{QCD} \cdot S_{26}^{\{(k)\}}$	Confined
10^{12}	$\frac{1}{3} \Lambda_{QCD} \cdot S_{26}^{\{(k)\}}$	Confined
1.5×10^{12}	0	Deconfined
$> T_c$	0	QGP

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_969 — Expanded 26D Ramanujan $S_{26}^{\{k\}}$
3. PAPER_970 — QGP Vacuum Density ρ_{QGP}
4. Clay Mathematics Institute — Yang-Mills and Mass Gap
5. Particle Data Group — Λ_{QCD} measurement

Cross-Links

Related Paper Relationship
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PAPER_969 $S_{26}^{\{k\}}$ acceleration factor
PAPER_970 QGP vacuum density (companion)
PAPER_972 ALICE centrality (experimental context)
PAPER_530 Millennium Hub (YM+Riemann+PvsNP)

<!-- PKG-YM-S225 -->

Session 225 Phonon-Physics Upgrade: Yang-Mills BCS Phonon Mass Gap

> Upgrade from PAPER_1318 (Yang-Mills Mass Gap via SCm BCS Phonon) and
 > PAPER_1070 (Yang-Mills Mass Gap VDS Bridge). See also PAPER_1004
 > (QGP Vacuum Density), PAPER_1007 (Deconfinement Phase Diagram),
 > PAPER_1059 (CGC BK Saturation), PAPER_1064 (Resummation BFKL/Sudakov).

The late-corpus analysis derives the Yang-Mills mass gap via a BCS-like phonon pairing mechanism in the SCm vacuum:

$$\Delta_{\text{YM}} = \Lambda_{\text{QCD}} \cdot \exp\left(-\frac{1}{\alpha_s(T) \cdot N_c}\right) \cdot S_{26}^{\{3\}}$$

where the running coupling evolves as:

$$\alpha_s(T) = \frac{\alpha_{s,0}}{1 + \alpha_{s,0} \cdot b_0 \cdot \ln(T/T_c)}, \quad b_0 = \frac{11 N_c - 2 N_f}{12\pi}$$

Physical mechanism: The SCm phonon field ($\omega_{\text{SCm}} = 1.25 \text{ THz}$) provides a pairing interaction analogous to the BCS electron-phonon coupling in superconductors. Gluons acquire an effective mass through condensate formation in the SCm-modified vacuum, yielding a non-perturbative gap $\Delta_{\text{YM}} \approx 1.736 \text{ GeV}$ (PAPER_1318 integer-primitive closure; lattice QCD anchor 1.7 GeV; supersedes 1.736 GeV registry-bug value).

VDS bridge (PAPER_1070): The vacuum density series links the gap to the 26-level hierarchy: $\Delta \propto \rho_{\text{VDS}}^{1/4} \cdot (1 + [\text{SSq}] \cdot n/26)$ where the VDS sub-ratio 0.108 places confinement in the sub-threshold regime.

QGP transition (PAPER_1004/1007): At $T > T_c \approx 170 \text{ MeV}$, the phonon coupling weakens ($\alpha_s \rightarrow 0$) and the gap closes, reproducing the deconfinement phase transition observed at ALICE/LHC.

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
> PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
> (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{-\kappa_i n/26}} \right]$$

where $(a)_n = a(a+1) \cdots (a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{S_{26}^{(3)}}{e^{-\kappa_i n/26}} \right]$$

This sum converges absolutely for $|S_{26}^{(3)}| < 1$ (satisfied by $|S_{26}^{(3)}| = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $S_{26}^{(3)} \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
Λ_{QCD}	—	217 MeV (0.217×10^9 eV)	QCD scale
T_c	—	1.5×10^{12} K	Deconfinement
$S_{26}^{(3)}$	—	0.57	String coupling
Gap at $T=0$	$\Delta_{\text{YM}}(0)$	$\Lambda_{\text{QCD}} \cdot S_{26}^{(k)}$	Confinement

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Mass gap existence	$\Delta_{\text{YM}} > 0$ for $T < T_c$	Confirmed
Gap closure	At $T = T_c$	Continuous
Λ_{QCD}	217 MeV	PDG consistent

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification **Sector:** Yang-Mills Confinement (Mass Gap)

§A.2 Core Equation $\boxed{\Delta_{\text{YM}}(T) = \Lambda_{\text{QCD}} \cdot \left(1 - \frac{T}{T_c}\right) \cdot S_{26}^{(k)}}$

§A.3 Lagrangian Contribution $\mathcal{L}_{\text{YM}} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} - V(\Delta_{\text{YM}}) + \frac{1}{2} \left(\frac{\partial \Delta_{\text{YM}}}{\partial T} \right)^2 \dot{T}^2$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\rightarrow$ vacuum density $\rightarrow S_{26}^{(k)} \rightarrow$ QCD confinement $\rightarrow$ Yang-Mills mass gap $\rightarrow$ quark masses

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS Δ_{YM} is proportional to $S_{26}^{(k)}$ — the VDS series directly determines the confinement scale.

§B.2 DVP Color confinement via flux tubes maps to DVP string modes in 26D.

§B.3 BSH Gap closure at T_c corresponds to BSH shell dissolution — buoyancy overcomes confinement.

§B.4 Summary

Metric	Value	Status
Λ_{QCD}	217 MeV	PDG 2024
T_c	1.5×10^{12} K	Lattice QCD
Gap	Closes at T_c	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1318	Yang-Mills Mass Gap via SCm BCS Phonon Coupling
PAPER_1006	ALICE Multiplicity SCm Phonon Scaling
PAPER_1007	Deconfinement Phase Diagram SCm Phonon Boundary
PAPER_1013	QGP ALICE Centrality $F_{U,Bi} dN/d\eta$ Scaling
PAPER_1059	Color Glass Condensate BK Saturation SCm
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1070	Yang-Mills Mass Gap VDS Bridge
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

11 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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